

i) Convert the expression $((a+b)+c*(d+e)+f)*(g+h)$ to
 1) prefix expression 2) postfix expression.

ii) Convert the infix expression

infix:

Step 1: prefix expression

→ +ab
 → +de
 → *c+de
 → ++ab*c+de
 → +++ab*c+def
 → +gh
 • Final multiplication:

prefix:

*+++ab*c+def+gh

Step 2: postfix expression

→ ab+
 → de+
 → cde+*
 → ab+cde+*+
 → Add f → ab+cde+*+f+
 → gh+

Final multiplication:

postfix:

ab+cde+*+f+gh+*

Two students obtain different postfix expressions for the same infix expression. Infix: $(A+B)*(C-D)/E$
 student A: $AB+CD-*E/$ student B: $AB+C*D-E/$
 Determine which postfix expression is correct and justify your answer by tracing the conversion process.

Given Infix:

Conversion:

1. $(A+B) \rightarrow AB+$

2. $(C-D) \rightarrow CD-$

3. $(A+B) * (C-D) \rightarrow AB+CD-*$

4. Divide by E $\rightarrow AB+CD- * E /$

Correct Postfix:

$$AB+CD-*E/$$

Therefore:

student A: $AB+CD-*E/$

Justification:-

student B performs multiplication between C and D before subtraction, which changes the original expression. The parentheses (C-D) must be evaluated first, so the correct postfix is:

$$AB+CD-*E/$$